

CHE 396 Quiz 4

Omar Takroui

TOTAL POINTS

88 / 100

QUESTION 1

11 88 / 100

- **0 pts** Correct

✓ - **5 pts** capital cost is wrong (only ISBL is taken into account)

✓ - **3 pts** Gross profit for first year of operation is wrong

- **10 pts** Depreciation is wrong

✓ - **2 pts** tax should be next year

- **5 pts** Discount factor is not right

- **0 pts** Click here to replace this description.

- **5 pts** Gross profit is wrong

- **5 pts** tax is wrong

- **10 pts** cash flow is wrong

- **10 pts** taxable income is wrong

- **10 pts** PV is wrong or not done

- **10 pts** NPV is not calculated or wrong

✓ - **2 pts** Depreciation is only taken for ISBL not full capital cost

- **100 pts** Not done or wrong

Omar Takroufi
Name

1. If ISBL cost of a plant is \$60,000 and annual fixed cost of production is \$15,000 and variable cost of production is \$10,000 and annual main product revenue is \$60,000, calculate the maximum 7 year NPV that can be achieved with a 15% interest rate and a 25% tax rate. Assume plant build in two years (capital paid in two years equally), and 50% operation the first year when operation starts. Use MACRS depreciation with a five-year recovery term. (MACRS depreciation charges for 5 year recovery, year 1 = 20, year 2 = 32, year 3 = 19.2, year 4 = 11.52, year 5 = 11.52 and year 6 = 5.76)

YEAR	Gross profit	Depreciation charges	Taxable income	Taxes paid	Cash flow	Discount factor	Present value of CF	NPV
0	-30,000	0	0	0	-30,000	1	-30,000	-30,000
1	-10,000	12000	2,000	500	-15,000	0.870	-13,050	-43050
2	35,000	19200	15800	3950	31,050	0.756	23,473	-19,577
3	35,000	11520	23480	5870	29,130	0.658	19,167	-410
4	35,000	6912	28088	7022	27,978	0.572	16,003	15,593
5	35,000	6912	28088	7022	27,978	0.497	13,905	29,498
6	35,000	3456	31544	7886	27,114	0.432	11,713	41,211
7	35,000	3456	31544	7886	27,114	0.376	10,195	51,406

$$\text{Depreciation} = \text{Total Investment} \times \left(\frac{\text{MACRS Factor}}{100} \right)$$

$$\text{Cash flow} = \text{Gross Profit} - \text{Taxes paid}$$

$$F_d = \frac{1}{(1 + \text{interest rate})^{\text{year}}}$$

$$\text{Present value of CF} = \text{Cash Flow} \times \text{Discount Factor}$$

\$51,406

Maximum 7-year
NPV